Time Restricted Eating in Alzheimer's Disease (TREAD): A Pilot Study

BARROW Neurological Institute

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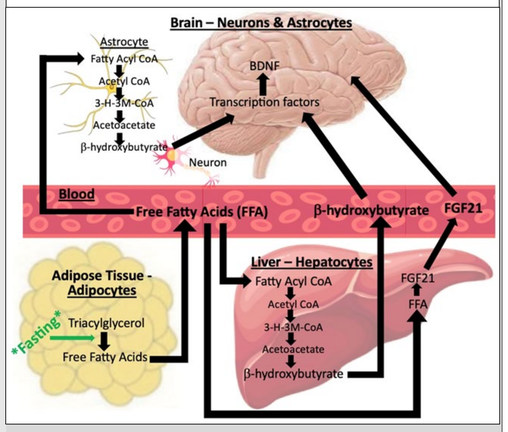
BACKGROUND

- **Time-restricted eating (TRE)** may impact cognition favorably
- Fasting periods result in ketone production and metabolic switching from glucose to fatty acid and ketone utilization, leading to a cascade of adaptive metabolic and cellular responses (**Figure 1**)
- Cortical glucose uptake is decreased in Alzheimer's disease (AD); ketone uptake is not affected
- Little is known about the potential benefits of TRE in patients with **mild cognitive impairment (MCI)**

Figure 1. Mechanistic model for TRE



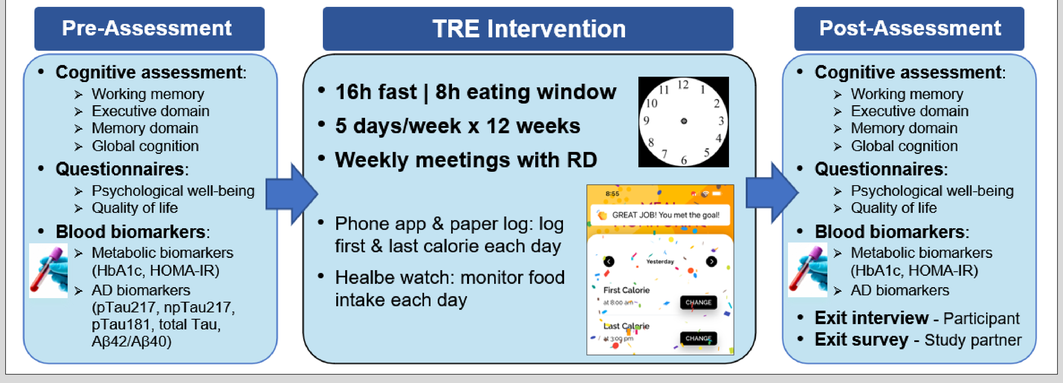
AIMS

- Investigate the feasibility of implementing a TRE intervention in patients with MCI
- Assess the impact of TRE on AD biomarkers, metabolic biomarkers, and cognition

METHODS

- **Study location**: Barrow Neurological Institute
- **Eligibility**: age 55-89y, Mayo Clinic Criteria for MCI, BMI >18.5 and <40.0 kg/m^2, supportive study partner to facilitate study visits and intervention activities

Figure 2. Study design



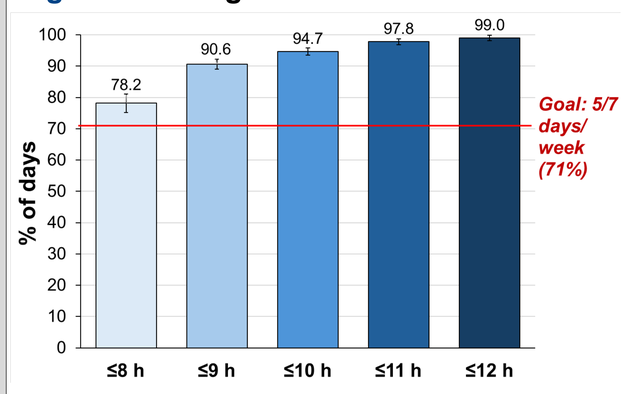
PRELIMINARY RESULTS

- N = 27 completed participants (17 men, 10 women)

Table 1: Baseline Characteristics

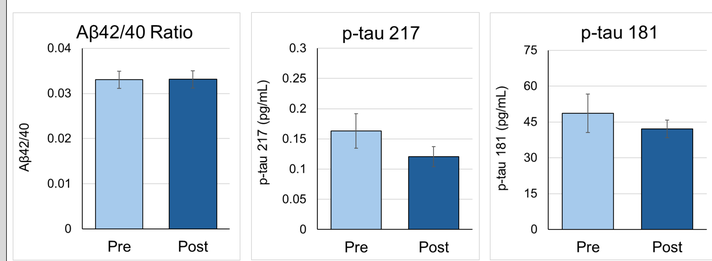
	Mean ± SEM		Range	
Age (y)	72.1 ± 1.1		63y - 82y	
BMI (kg/m^2)	27.3 ± 0.9		18.6 - 34.0	
MMSE score	25.3 ± 0.7		18-30	
Trail Making A	Raw score	T score	Raw score	T score
	71.1±7.2	42.4±2.4	25-198	18-65
Trail Making B	Raw score	T score	Raw score	T score
	114.6±9.9	40.4±2.1	50-240	18-59

Figure 3. Eating Windows Achieved



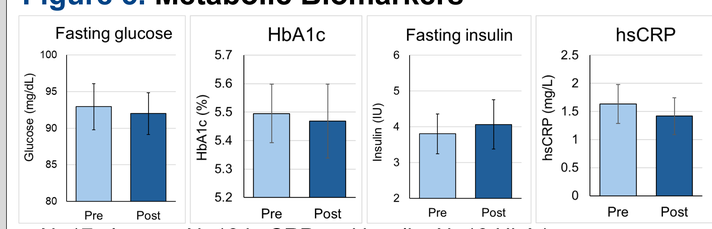
- Participants were highly adherent
- 8h eating window achieved on 78.2% of all days
- 10h eating window achieved on 94.7% of all days

Figure 4. AD Biomarkers (plasma)



- N=19 for Aβ42/40 ratio, N=21 for p Tau 181 and p Tau 217
- Bars reflect mean + SEM

Figure 5. Metabolic Biomarkers



- N=17 glucose, N=18 hsCRP and insulin, N=19 HbA1c
- Bars reflect mean + SEM

Table 2: Exit Interview

Question	% of Participants
Participant believed that they benefitted from study participation	93%
Participant believed they could incorporate TRE long term	95%
Participant felt that the 16-hour fasting window was not too long	74%
Participant would recommend this study	85%

DISCUSSION

- Preliminary observations from the TREAD trial indicate high feasibility and adherence to a TRE intervention among patients with MCI.
- Completion of the remaining participants will shed light on the potential impact of TRE on cognition and blood biomarkers of AD.

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